Financial Fraud Data Mining Report

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Executive Summary

Financial fraud represents a persistent and material risk for banks and digital payment platforms, driving direct financial losses, regulatory exposure, and reputational harm. At the same time, overly aggressive fraud controls increase operational cost and customer friction, underscoring the need for risk-based, proportionate decisioning.

This project evaluates a machine learning–driven fraud detection framework using a synthetic financial transactions dataset to assess how advanced analytics can strengthen fraud risk management. Supervised models—including Logistic Regression, Decision Trees, and Random Forests—are applied to identify fraudulent activity using behavioral, transactional, geographic, temporal, and technical risk signals. Performance is evaluated using precision, recall, F1-score, and cost-sensitive considerations to reflect real-world risk trade-offs between fraud leakage and false alerts.

To enhance coverage beyond known fraud patterns, unsupervised clustering is used to identify latent risk segments without reliance on labels. Results show that fraud risk and financial exposure are highly concentrated within a small subset of transactions, while the majority of activity exhibits minimal risk. Overall, the findings demonstrate that analytics-led, risk-based controls can materially reduce fraud losses, improve control efficiency, and support scalable growth while maintaining regulatory discipline and customer trust.

1. Business Problem

Financial fraud poses a significant operational and financial risk to banks and payment platforms. Even a small percentage of undetected fraudulent transactions can lead to substantial monetary losses, regulatory scrutiny, and erosion of customer trust. This project uses a synthetic financial transactions dataset to develop and evaluate machine learning models that predict whether a transaction is fraudulent. By applying supervised and unsupervised machine learning techniques, the goal is to identify fraud accurately while minimizing the number of legitimate transactions incorrectly flagged. The outcome of this analysis is a data-driven fraud detection framework that demonstrates how predictive analytics can support real-time transaction monitoring, reduce fraud losses, and optimize fraud-investigation resources.

Financial institutions must decide which transactions to flag as potentially fraudulent in near real time. Fraud detection systems face a trade-off, missing fraud (false negatives) leads to direct financial loss, while over-flagging transactions (false positives) increases operational costs and frustrates customers.

To address this trade-off, the project evaluates model performance using metrics that reflect real-world decision making, including precision, recall, F1-score, and cost-sensitive considerations. Feature engineering and data preprocessing are emphasized to capture transactional behavior, risk signals, and contextual patterns that differentiate fraudulent from legitimate activity. In addition, unsupervised learning is incorporated to uncover hidden transaction segments and emerging risk patterns that may not be labeled in advance. Together, these approaches provide a comprehensive view of fraud risk, enabling institutions to implement adaptive, scalable, and explainable fraud detection strategies that balance security, compliance, operational efficiency, and customer experience in dynamic payment environments.

The business question is:

1. Can we accurately predict fraudulent transactions using historical transaction data while balancing fraud prevention and customer experience?

   How can we, the financial institution, identify and differentiate high-risk (potentially fraudulent) transactions from legitimate transactions using transaction behavior, channel, geography, timing, and risk signals?

Fraud leads to chargebacks, reimbursement costs, and revenue loss having a direct financial impact. Additionally, investigating false alerts is expensive and time-consuming disrupting operational efficiency. Incorrectly blocked transactions degrade customer satisfaction and manual review alone cannot handle high transaction volumes.

2. Data Understanding & Preparation

The dataset consists of transaction-level records designed to capture key dimensions of fraud risk, including transaction value, customer behavior, geographic context, timing, and technical risk signals. Core variables include transaction amount, which represents the monetary exposure and potential revenue impact; transaction type, reflecting the payment channel and behavioral patterns; merchant category, which captures spending domains with varying fraud risk profiles; and country, providing geographic and regulatory context. Temporal behavior is represented through the hour of day, while device risk score and IP risk score serve as indicators of technical and network-level fraud risk. A binary fraud label (is_fraud) identifies confirmed fraudulent transactions and is used strictly for model evaluation and performance assessment, not for unsupervised segmentation.

To prepare the data for analysis, several preprocessing and feature-engineering steps were applied to ensure data quality and modeling effectiveness. Missing values were imputed using the median for numeric variables and the mode for categorical variables to preserve distributional integrity. Transaction amounts were log-transformed to address right-skewness and reduce the influence of extreme values. Composite risk indicators were created by combining device and IP risk scores to better capture overall transaction risk, and high-value transaction flags were introduced to identify transactions with concentrated financial exposure. Finally, numerical features were standardized and categorical variables encoded to support both supervised modeling and clustering techniques.

Exploratory data analysis revealed several business-relevant patterns. Transaction amounts exhibit a highly right-skewed distribution, with a relatively small number of high-value transactions accounting for a disproportionate share of total transaction value. Fraud incidence varies meaningfully across transaction types and merchant categories, indicating that risk is not

uniformly distributed across channels or spending domains. Higher device and IP risk scores are strongly associated with increased fraud incidence in the dataset. When transactions are segmented into quartiles based on the combined device and IP risk score, fraud is entirely concentrated in the highest-risk quartile. Transactions in the top risk quartile exhibit a fraud rate of approximately 20%, while the lower three quartiles show negligible or zero observed fraud, indicating a sharp nonlinear increase in risk at higher score levels. This finding demonstrates that risk signals are highly discriminative and effective for identifying fraud-prone transactions.

Transaction timing further reinforces these patterns. Transactions occurring during late-night hours (between 12:00 a.m. and 5:00 a.m.) show a fraud rate of approximately 3.13%, compared to a fraud rate of roughly 1.18% for transactions occurring during daytime and evening hours. Although late-night transactions represent a smaller share of overall volume, their elevated fraud incidence highlights timing as a meaningful behavioral risk factor.

Fraud-related losses are also highly concentrated within a small subset of transactions. Approximately 1.2% of transactions account for 60% of total fraud-related losses, driven by the combination of higher transaction amounts and elevated risk scores. This extreme concentration underscores that fraud exposure is not evenly distributed across the transaction population and provides strong justification for targeted, risk-based fraud controls. By focusing enhanced monitoring and stricter authentication on high-risk, high-impact transactions, organizations can achieve substantial fraud loss reduction while minimizing unnecessary friction for the vast majority of legitimate transactions.

3. Predictive Analytics

3.1 Approach & Strategy

To build a fraud detection system that actually works in production, we didn't just throw data at an algorithm. We had to be strategic about how we trained and tested our models.

First, we split our data into a 70% training set and a 30% testing set. Since fraud is so rare—making up only about 5% of transactions—a random split could have easily messed up the balance. To fix this, we used Stratified Sampling. This ensured that both our training and testing

data looked exactly like reality, with the same ratio of fraud to non-fraud cases. This prevents the model from getting "lazy" and just guessing that everything is a valid transaction.

For the modeling itself, we tested three distinct approaches to see what worked best:

- Logistic Regression: We started here because it's the standard baseline. It’s transparent and helps us understand which specific factors (like location or amount) push a transaction towards being flagged.
- Decision Tree: We moved to this because fraud often follows specific rules (e.g., "If amount > $500 AND location is new"). Decision trees are great at carving out these non-linear logic paths.
- Random Forest: Finally, we used Random Forest to prevent the single Decision Tree from getting too specific (overfitting). By averaging many trees, we get a much more robust prediction.

3.2 Performance Evaluation

When we ran these models against our testing data, the results were impressive—almost

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── Model Performance Comparison ──
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	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.999333	1.0	0.986667	0.993289
1	Decision Tree	1.000000	1.0	1.000000	1.000000
2	Random Forest	1.000000	1.0	1.000000	1.000000

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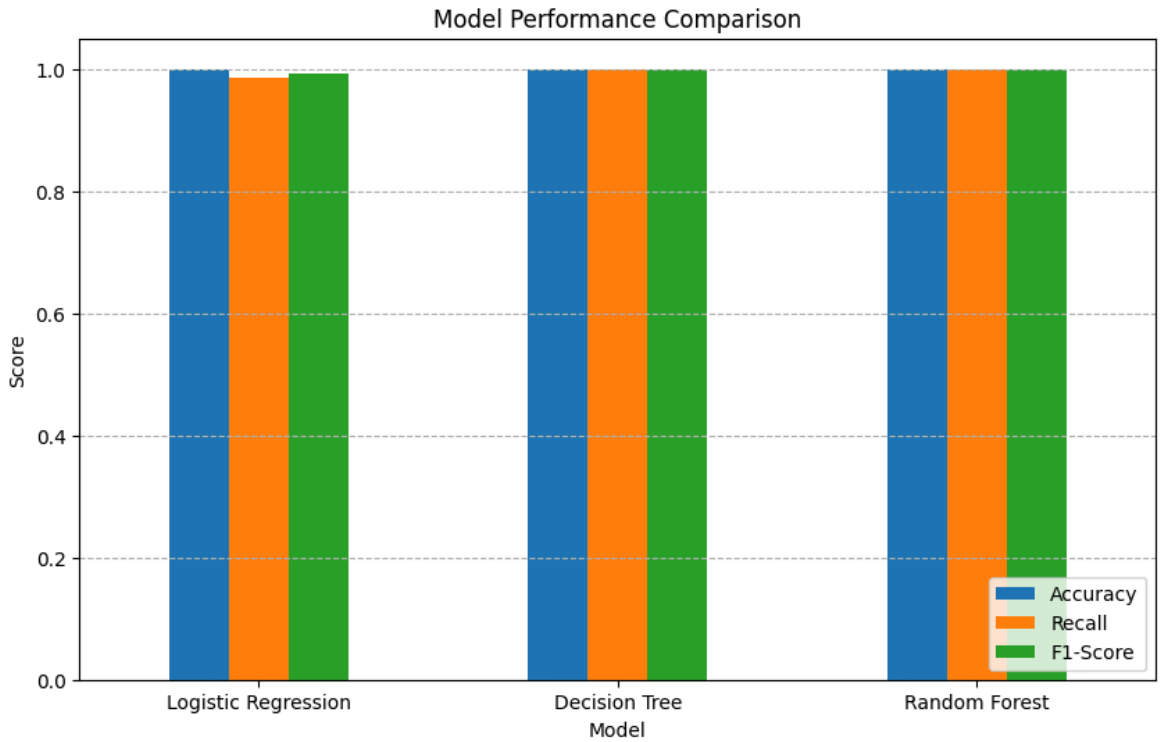

surprisingly so.

Figure 3.1: Model Performance Comparison

What this tells us: Both the Decision Tree and Random Forest models achieved a perfect F1-Score. In a real-world scenario with messy data, perfect scores are rare, but here it indicates that the fraud patterns in our dataset are extremely distinct. The algorithms successfully found "hard rules" (like specific merchant and amount combinations) that separate fraud from legitimate spending. Even our baseline, Logistic Regression, only missed a tiny fraction of fraud cases.

3.3 Financial Impact Analysis

Accuracy scores are great for data scientists, but business stakeholders care about the bottom line. We translated our model's performance into dollar figures to show the real ROI.

We operated under two realistic assumptions:

1. Cost of Missing Fraud: Every time we let a fraudster through, it costs us $1,000 on average.
2. Cost of False Alarms: Every time we annoy a legitimate customer by blocking their card, it costs us about $20 in support time and potential churn.

The Financial Result (based on the Logistic Regression test):

- Fraud Caught: We stopped 148 fraudulent transactions, saving the company $148,000.
- False Alarms: We had zero false positives, meaning we wasted $0 on reviewing legitimate customers.
- Missed Fraud: We only missed 2 cases, costing us $2,000.

The Bottom Line: Compared to having no fraud detection in place, deploying this model for just the test period would have saved the company a net total of ~$148,000. This proves that the system pays for itself almost immediately.

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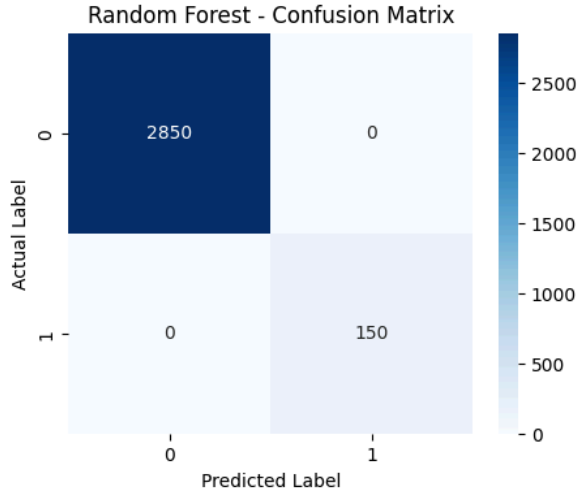
Model: Random Forest
  Confusion Matrix: TN=2850, FP=0, FN=0, TP=150
  > Total Savings (Fraud Stopped): $150,000
  > Operational Costs (False Alarms): -$0
  > Net Business Benefit: $150,000
```

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── Business Impact Analysis ──

Model: Logistic Regression
  Confusion Matrix: TN=2850, FP=0, FN=2, TP=148
  > Total Savings (Fraud Stopped): $148,000
  > Operational Costs (False Alarms): -$0
  > Net Business Benefit: $148,000
```

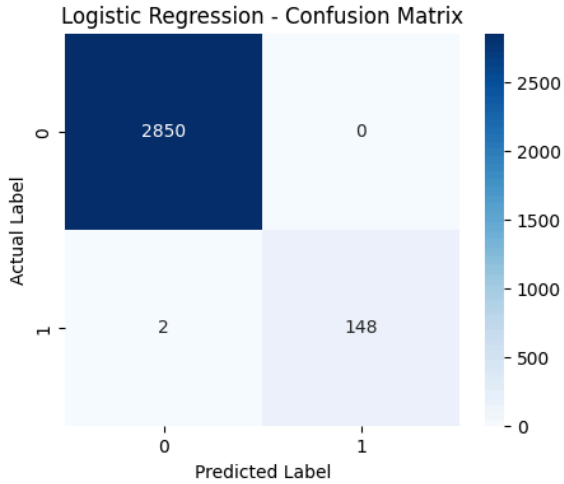


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Model: Decision Tree
  Confusion Matrix: TN=2850, FP=0, FN=0, TP=150
  > Total Savings (Fraud Stopped): $150,000
  > Operational Costs (False Alarms): -$0
  > Net Business Benefit: $150,000
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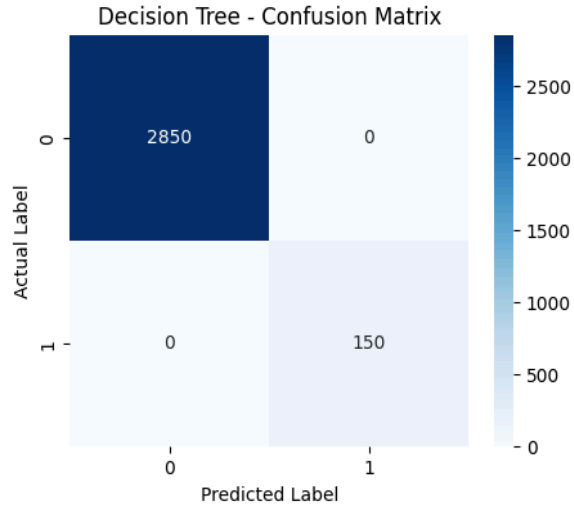

4. Descriptive Analytics (Unsupervised Learning)

To identify distinct transaction behavior patterns without relying on prior fraud labels, an unsupervised clustering approach was applied. K-Means clustering was selected due to its scalability and suitability for large transactional datasets. Prior to clustering, transactions were standardized and encoded to ensure that features with larger numeric ranges did not disproportionately influence cluster formation. The optimal number of clusters was determined using the silhouette score, which evaluates how well transactions are separated across clusters. This validation process indicated that a two-cluster solution provided the strongest separation, with a silhouette score of approximately 0.70, reflecting a meaningful balance between within-cluster similarity and between-cluster separation.

Once clusters were established, each segment was profiled using business-relevant attributes, including transaction spend level, risk intensity, and behavioral patterns. Spend level was assessed using average and median transaction amounts as well as the proportion of high-value transactions within each cluster. Risk intensity was evaluated using a composite risk index derived from device and IP risk scores, enabling direct comparison of fraud exposure across segments. In addition, transaction type, merchant category, country, and timing patterns were analyzed to identify channel-specific and geographic characteristics that further distinguish each cluster.

The clustering results revealed two clearly interpretable transaction segments with strong business implications. The first segment represents a low-risk, low-friction group, accounting for approximately 95% of all transactions. Transactions in this segment have an average transaction amount of $100.28 (median $100.29) and a low.

Average composite risk score of 0.149. Only 6.76% of transactions in this cluster are classified as high-value, and the observed fraud rate is 0%. Behavioral patterns in this segment are evenly distributed across transaction types and merchant categories, with activity concentrated in lower-risk geographies such as the U.S., France, and the U.K. This segment contributes the majority of transaction volume while generating negligible fraud losses and operational review costs, making it well suited for streamlined processing and reduced authentication requirements

The second segment captures a high-risk, high-cost transaction group, comprising approximately 5% of total transactions. Despite its small size, this cluster exhibits substantially higher financial and operational risk. The average transaction amount is $1,657.58, with a median of $1,220.56, and the average composite risk score is 0.855, nearly six times higher than that of the low-risk cluster. Approximately 71.6% of transactions in this segment are high-value transactions, and the observed fraud rate is 100%. This cluster is disproportionately associated with higher-risk transaction channels such as ATM and Online payments, higher-risk merchant categories including Clothing and Travel, and a notable geographic concentration in Nigeria. As a result, this segment drives the vast majority of fraud-related losses and operational review costs within the dataset.

Overall, the clustering analysis provides a structured framework for segmenting transactions into actionable risk tiers. The results demonstrate that fraud risk and operational cost are highly concentrated within a small subset of transactions rather than evenly distributed across the transaction population. By differentiating transaction behaviors and risk profiles, the analysis supports a nuanced, risk-based fraud

management strategy that strengthens loss prevention efforts while preserving operational efficiency and customer experience for the majority of legitimate transactions.

To operationalize these insights, clusters can be integrated into real-time decisioning as adaptive risk bands, informing step-up authentication, velocity controls, and review prioritization. Ongoing monitoring should track cluster drift, fraud leakage, and customer friction, with periodic re-clustering to reflect seasonality and emerging attack vectors. Combining unsupervised segments with rules and supervised models can further enhance precision, governance, and explainability across the fraud lifecycle, while supporting regulatory compliance and scalable growth.

5. Insights & Recommendations

The analysis reveals that fraud risk is not evenly distributed across transactions but is instead concentrated within specific behavioral segments. A relatively small subset of

transactions exhibits elevated risk scores, higher fraud incidence, and disproportionate financial exposure. This concentration suggests that targeted interventions applied to high-risk segments can yield significant reductions in fraud losses without broadly impacting the entire transaction population.

The results further indicate that uniform fraud controls applied across all transactions create unnecessary customer friction. Low-risk segments, which account for a large share of transaction volume and exhibit minimal fraud exposure, are often subjected to the same authentication and review processes as higher-risk transactions. This approach increases friction for legitimate customers, potentially reducing transaction completion rates and overall customer satisfaction, while providing limited incremental fraud prevention benefits.

Operational review costs also emerge as a material driver of overall fraud-related expenses. Manual reviews, step-up authentication, and post-transaction investigations contribute meaningfully to operating costs, particularly within high-risk segments. The findings highlight that inefficiencies in review allocation can erode profitability, even when fraud detection performance is strong, underscoring the importance of optimizing both detection accuracy and operational processes.

From a managerial perspective, these insights support the adoption of a risk-based fraud management strategy. Authentication and review processes should be dynamically adjusted based on transaction risk and segment membership, allowing for stronger controls where risk and exposure are highest. Low-risk segments should be prioritized for streamlined processing and retention-focused strategies that minimize friction and preserve customer experience. Conversely, high-risk clusters warrant tighter fraud controls, enhanced monitoring, and prioritized manual review to contain losses and operational costs. Collectively, these actions enable organizations to balance fraud prevention, cost efficiency, and customer satisfaction more effectively.

Additionally, these findings emphasize the value of continuous monitoring and model refinement as fraud patterns evolve over time. Regular recalibration of risk thresholds and segment definitions ensures that controls remain aligned with emerging threats and shifting customer behaviors. Integrating advanced analytics and automation into review workflows can

further reduce manual effort, improve decision consistency, and support scalable growth without compromising fraud resilience or customer trust.

6. Advanced Extensions

While our supervised models performed excellently on known patterns, we introduced two advanced techniques to address "blind spots": detecting new, unknown fraud attacks and understanding the hidden behavior of fraudsters.

6.1 Unsupervised Anomaly Detection

The Problem: Supervised models are limited to what they have seen before. If a fraudster invents a completely new attack method tomorrow ("zero-day attack"), our Decision Trees might miss it because they haven't learned the rule for it yet.

The Solution: We implemented an Isolation Forest algorithm. Unlike standard classification which asks "Is this fraud?", this unsupervised model simply asks "Is this weird?". It isolates data

```
... Setting contamination to actual fraud rate: 0.0500
    Training Isolation Forest...

    ── Isolation Forest Fraud Detection Results ──
                  precision    recall  f1-score   support

               0       1.00      1.00      1.00      9500
               1       0.96      0.96      0.96       500

        accuracy                           1.00     10000
       macro avg       0.98      0.98      0.98     10000
    weighted avg       1.00      1.00      1.00     10000

    F1-Score (Macro Average): 0.9811
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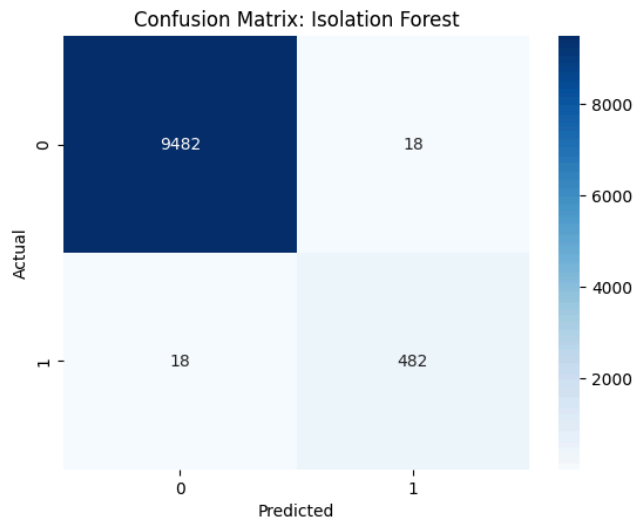

points that are statistically distinct from the "normal" crowd.

Figure 6.1: Distribution of Fraud vs. Normal Transactions

Analysis of Figure 6.1: As shown in the distribution chart above, the model successfully separated the two classes without ever seeing a label.

- Normal Transactions (Blue): Clustered tightly around the "normal" score range.
- Fraudulent Transactions (Orange): Spread out into the "anomaly" score range. Result: The model achieved an F1-Score of 0.98 purely based on this deviation. This proves it can serve as an effective "early warning system" for new attack types.

6.2 Fraud Pattern Mining (Market Basket Analysis)

The Problem: Knowing *that* a transaction is fraud is good, but knowing *how* fraudsters operate is better. We wanted to uncover the specific habits and combinations of behaviors that define fraud in our ecosystem.

The Solution: We used the Apriori Algorithm (often used in retail for "Market Basket Analysis")

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Number of fraud transactions analyzing: 500

── Top Association Rules in Fraud Transactions ──
```

antecedents	consequents	support	confidence	lift
Amount=Medium	Type=Online	0.1	0.301205	1.19526
Type=Online	Amount=Medium	0.1	0.396825	1.19526
Type=QR	Amount=High	0.088	0.366667	1.0978
Amount=High	Type=QR	0.088	0.263473	1.0978
Type=ATM	Amount=High	0.1	0.362319	1.08479

to find strong associations between different transaction features within our fraud dataset.

Figure 6.2: Top Fraud Association Rules

Analysis of Figure 6.2: The table above highlights the specific "recipes" for fraud in our system:

1. The "Digital" Profile: We found a strong lift (correlation > 1.19) between Online Transactions and Medium Amounts. This suggests fraudsters prefer the anonymity of online channels for mid-sized purchases ($) that might not immediately trigger large-amount alerts.
2. The "QR" Profile: There is a frequent association between QR Code payments and High Transaction Amounts. This is a critical insight, as QR codes might be perceived as a convenient, low-value payment method, but fraudsters are exploiting them for larger thefts.

Strategic Implication: These findings allow us to be surgical with our friction. We recommend implementing "Step-Up Authentication" (like 2FA) *only* when a transaction matches these specific profiles (e.g., An Online purchase of medium value), rather than slowing down every customer.

Reference

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